

React HandsOn 8:-

Theory:

- Explain react state.

React State is a built-in object in React that is used to store and manage data that belongs to a component. It contains information that can change over time, such as user input, button clicks, form values, API responses, or counters. Unlike props, which are passed from a parent component and are read-only, state is managed within the component itself and can be updated whenever required. React automatically re-renders the component whenever its state changes, ensuring that the user interface always displays the latest data.

In class components, state is initialized inside the constructor using the `this.state` object. To update the state, React provides the `setState()` method. Using `setState()` ensures that React detects the change and re-renders the component with the updated values. Directly modifying the state (for example, `this.state.count++`) is not recommended because React will not recognize the change correctly.

In functional components, state is managed using the `useState()` hook. The hook returns the current state value and a function to update it. This is the modern and preferred approach in React applications.

Key Features of React State

- Stores data that belongs to a component.
- Used to manage dynamic and changing data.
- Initialized in the constructor (class components) or using `useState()` (functional components).
- Updated using `setState()` (class components) or the setter function returned by `useState()`.
- Any change in state automatically re-renders the component.
- State is private to the component and cannot be accessed directly by other components unless passed as props.

Practical:

Code:

```
1 import React, { Component } from "react";
2
3 class CountPeople extends Component {
4   constructor(props) {
5     super(props);
6     this.state = {
7       entryCount: 0,
8       exitCount: 0
9     };
10  }
11  UpdateEntry = () => {
12    this.setState({
13      entryCount: this.state.entryCount + 1
14    });
15  };
16  UpdateExit = () => {
17    this.setState({
18      exitCount: this.state.exitCount + 1
19    });
20  };
21  render() {
22    return (
23      <div>
24        <h1>Mall Entry Counter</h1>
25        <h2>People Entered : {this.state.entryCount}</h2>
26        <button onClick={this.UpdateEntry}>
27          Login
28        </button>
29        <br />
30        <h2>People Exited : {this.state.exitCount}</h2>
31        <button onClick={this.UpdateExit}>
32          Exit
33        </button>
34      </div>
35    );
36  }
37 }
38 export default CountPeople;
```

```
JS Cart.js U JS CountPeople.js U JS App.js M X # App.css M JS index.js M
1 import CountPeople from "../CountPeople";
2 import "../App.css";
3
4 function App() {
5   return (
6     <div className="App">
7       <CountPeople />
8     </div>
9   );
10 }
11 export default App;
12
```

```
JS Cart.js U JS CountPeople.js U JS App.js M # App.css M X JS index.js M
1 .App {
2   text-align: center;
3   margin-top: 60px;
4   font-family: Arial, Helvetica, sans-serif;
5 }
6 button {
7   width: 120px;
8   padding: 10px;
9   margin: 10px;
10  font-size: 16px;
11  cursor: pointer;
12 }
13 h1 {
14   color: navy;
15 }
16 h2 {
17   color: green;
18 }
```

```
JS Cart.js U    JS CountPeople.js U    JS App.js M    # App.css M    JS index.js M X
1  import React from "react";
2  import ReactDOM from "react-dom/client";
3  import "./index.css";
4  import App from "./App";
5
6  const root = ReactDOM.createRoot(document.getElementById("root"));
7  root.render(
8    <React.StrictMode>
9      <App />
10    </React.StrictMode>
11  );
12
```

Output:

Mall Entry Counter

People Entered : 0

Login

People Exited : 0

Exit

Mall Entry Counter

People Entered : 4

Login

People Exited : 2

Exit